

Movie Recommendation System with Collaborative Filtering

PHASE 4 PROJECT

Project Objective

In this project, we will build a collaborative filtering recommender system using surprise library. Some of the key highlights of this project are:

1. Use surprise's built-in reader class to process data to work with recommender algorithms
2. Obtain a prediction for a specific user for a particular item
3. Introduce a new user with rating to a rating matrix and make recommendations for them
4. Create a function that will return the top 5 movie recommendations for a user, based on their ratings of other movies

Business and Data Understanding

- ▶ We will be making a movie recommendations based on the MovieLens <https://grouplens.org/datasets/movielens/latest/> dataset from the GroupLens research lab at the University of Minnesota.
- ▶ The MovieLens dataset contains the following files; links.csv, movies.csv, ratings.csv and tags.csv.
- ▶ For this project, we will only focus on the movies.csv and ratings.csv where we will get userId, movieId, rating and title(movie) columns that will be necessary for our recommender system.

Modelling

- ▶ We will determine the best model using the Python library called surprise. Surprise has various algorithms that can help us try out multiple different types of collaborative filtering engines ranging from Memory/Neighborhood based methods and Model based method.
- ▶ We will use RMSE (Root Mean Squared Error) to measure how far off our predictions are from the actual values. Lower values of RMSE indicate that the model's predictions are closer to the actual values, meaning the model is more accurate.
- ▶ The model with the lowest RMSE will then be used to make predictions and recommendation.

Predicting and Making Recommendations

The process of making the recommendations will as follows:

1. Create a list of user rating for a new user
2. Make predictions with new ratings
3. Order the predictions from highest to lowest rated
4. return the top 5 recommendations

Deployment/Conclusion

- ▶ We have been able to implement a movie recommender system with collaborative filtering based on the Singular Value Decomposition (SVD) model that can be used to predict and make recommendation of the top ranked movies..